Per-Op Gradient Subspaces:
A Diagnostic for Linear-Centroid Coupling in Multitask Training

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April 28, 2026

What the paper does. The paper introduces a parameter-space diagnostic for feature formation in deep networks. Given a network and a task, the diagnostic computes per-task loss gradients $g_{\text{op}}(t) = \nabla_\theta L_{\text{op}}|_{\theta_t}$ at a series of training checkpoints, takes a rolling-window SVD of these gradients, and measures how strongly perturbations along the top right singular vectors v_k move the input-space centroids $\mu_x = \nabla_x \ell_y(x)$ defined by the Linear Centroids Hypothesis (LCH) of Walker et al. [1]. The result is a quantitative bridge between an optimisation object (the gradient SVD basis) and a representational object (LCH features), applicable in single-task and multitask training.

Diagnostic. At each checkpoint t, for a unit direction v_k in attention-parameter space,

$$R_k(t) \;=\; \frac{\text{mean}_x \, \|\mu_x(\theta_t + \varepsilon v_k) - \mu_x(\theta_t - \varepsilon v_k)\|^2}{\text{median}_j \;\; \text{(same with random } r_j)}.$$

$R_k(t)$ measures how much the centroid manifold moves along v_k, normalised by the response to a random parameter direction. We compute R_k along directions extracted from two candidate sources — the rolling SVD of the AdamW update step $\Delta\theta_t$ (*update SED*) and the rolling SVD of the loss gradient $g(t)$ (*gradient SED*) — under matched window W, rank K, perturbation ε, and probe set.

Main empirical results. **(1) Single-task** (4 ops, 3 seeds; cf. Figure 2 of the main paper). Per-op gradient SED gives $\bar{R}_k$ peaks of 100–330 across addition, subtraction, multiplication, and a^2+b^2 mod 97. A window-size ablation (Table 4 of the main paper) shows that the dominant contribution comes from the instantaneous gradient direction itself ($R_1 = 648$ at W=1, decaying smoothly to 140 at W=40); the rolling-window SVD acts as a denoising mechanism. The signal is robust to K and probe-set seed within $\sim 5\%$ and remains in the linear-response regime for $\varepsilon_{\text{rel}} \leq 0.005$.

(2) Multitask (4-head shared encoder, all 4 ops trained jointly to grokking). Per-task gradient SED, applied to each op's own gradient $g_{\text{op}}(t)$, gives $\bar{R}_k = 20$–45 across all four ops:

op	per-op gradient SED ($\bar{R}_k$ peak)
add	**31.5**
sub	**44.9**
mul	**23.8**
sq	**21.8**

Per-task gradient SVD is the operative tool here: it recovers each op's feature-formation subspace from a shared training trajectory, distinguishing per-task circuit formation in the shared encoder.

(3) Causal intervention (3 seeds, 5 conditions, on add and mul; gradient-projected variant). Constraining the attention-parameter gradient to a rank-3 subspace at each step:

condition	grok step (mean)	speedup vs. control
control (full-rank gradient)	3358	1.00×
remove SED rank-3	3300	1.02×
keep only SED rank-3	**1525**	**2.20×**
remove random rank-3	3358	1.00×
keep only random rank-3	**1425**	**2.36×**

Constraining the attention update to any rank-3 subspace accelerates grokking by $\approx 2.3\times$. The speedup is rank-specific: SED-derived and random rank-3 are within seed variation. The natural full-rank attention update is therefore *rank-redundant* for the grokking transition under our hyperparameters — only a few effective parameter directions are needed at each step. The same pattern reproduces on multiplication (15 additional runs).

Interpretation. Together the three results characterise the per-task gradient subspace as a measurement object and a structural one. As a measurement object: top right singular vectors of $g_{\mathrm{op}}(t)$ are the parameter directions along which an op's centroid manifold is most sensitive at time t, and the resulting R_k values quantify this sensitivity. As a structural object: attention-update progress on grokking lives in a low-rank (≈ 3) parameter subspace, and constraining the optimiser to that rank accelerates convergence without loss of accuracy.

Methodological cost and scope. The diagnostic adds one backward pass per task per measurement checkpoint, cheap relative to training. The single-task correspondence between gradient direction and LCH features is near-tautological by the chain rule (both are built from network Jacobians); the multitask use of per-task SVD is the mechanistically substantive case. At LLM scale the $(N \times P_{\mathrm{attn}})$ gradient matrix is too large to materialise; a Lanczos approximation of the empirical Fisher's top eigenvectors recovers the same rank-K basis through Hessian-vector products and is the natural extension.

Scope of claims. All experiments use a 2-layer pre-norm Transformer with d_{model}=128 on mod 97 modular arithmetic, AdamW with β_2=0.98 and weight decay 1.0. Three random seeds on each single-task op, three on each intervention condition. Generalisation to LM-scale multitask training and other optimisers is open.

Code, training caches, and per-checkpoint data: `https://github.com/skydancerosel/grokking-integrability`

References

[1] T. Walker, A. I. Humayun, R. Balestriero, R. Baraniuk. *The Linear Centroids Hypothesis: How Deep Network Features Represent Data.* arXiv:2604.11962, 2026.